Retail Inventory Analytics — Project 2

Challenge: RETAIL — WHICH PRODUCTS SHOULD THE STORE STOCK MORE? (DS_Day01_04)

Dataset note: the supplied challenge PDF provides the schema and tasks but no row-level dataset. This implementation uses synthetic data for demonstration. The PDF also does not specify an explicit prediction label, so Fast_Moving is transparently derived from the 75th percentile of Units_Sold.

Key metrics

Metric	Value
Rows after cleaning	2400
Fast-moving threshold	39.00 units
Stock-out rate	8.75%
Mean turnover	0.403
Promotion avg units	35.23
No-promotion avg units	26.70
RF Accuracy	0.717
RF Precision	0.411
RF Recall	0.244
RF F1	0.306

Key insights

• Category sales identify the main drivers of unit movement.

• Promotion performance should be compared with baseline sales before increasing promotional inventory.

• Store stock-out rates identify locations requiring replenishment review.

• Inventory turnover highlights products converting stock into sales quickly.

• High-turnover or high-stock-out products are candidates for higher replenishment levels.

• Low-turnover and low-stock-out products are candidates for tighter inventory levels.

• Random Forest predictions use prior product demand as a historical signal without using the current day's sales.

Practical action plan

• Increase replenishment attention for high-turnover or high-stock-out products.

• Review low-turnover products for overstock risk.

• Compare promotion uplift by category and store before allocating extra inventory.

• Use store-level stock-out rates to rebalance inventory.

• Re-evaluate model predictions periodically as demand patterns change.